

Practical 2 (Task 1)

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varshini@DESKTOP-BLT5L3R:~$ nano fork_exec.c
varshini@DESKTOP-BLT5L3R:~$ gcc fork_exec.c -o fork_exec
varshini@DESKTOP-BLT5L3R:~$ ./fork_exec
Enter a Linux command: ls

Parent Process
Parent PID : 1274
Child PID  : 1275

Child Process
Child PID : 1275
Parent PID: 1274

Executing command: ls
OSSP  fork_exec  fork_exec.c  hello  hello.c  process.c

Child process completed.
varshini@DESKTOP-BLT5L3R:~$ |
```

```
GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    char command[100];
    pid_t pid;

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }

    if (pid == 0) {
        // Child process
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        printf("\nExecuting command: %s\n", command);

        execlp(command, command, NULL);

        // Executes only if execlp() fails
        perror("exec failed");
        exit(1);
    }
    else {
        // Parent process
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("\nChild process completed.\n");
    }

    return 0;
}
```

Practical 2 (Task 2)

```
varshini@DESKTOP-BLTSL3R:~$ uname -a
Linux DESKTOP-BLTSL3R 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 GNU/Linux
varshini@DESKTOP-BLTSL3R:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      12
On-line CPU(s) list:        0-11
Vendor ID:                   GenuineIntel
Model name:                  13th Gen Intel(R) Core(TM) i5-1334U
CPU family:                  6
Model:                      186
Thread(s) per core:         2
Core(s) per socket:         6
Socket(s):                   1
Stepping:                    3
BogoMIPS:                    4992.00
Flags:                       fpu vme de pse tsc msr pae mce cx8 a
pic sep mtrr pge mca cmov pat pse36
clflush mmx fxsr sse sse2 ss ht sysc
all nx pdpe1gb rdtscp lm constant_ts
c rep_good nopl xtopology tsc_reliab
le nonstop_tsc cpuid tsc_known_freq
pni pclmulqdq vmx ssse3 fma cx16 pci
d sse4_1 sse4_2 x2apic movbe popcnt
tsc_deadline_timer aes xsave avx fl6
c rdrand hypervisor lahf_lm abm 3dno
wprefetch ssbd ibrs ibpb stibp ibrs_
enhanced tpr_shadow ept vpid ept_ad
fsgsbase tsc_adjust bmi1 avx2 smep b
mi2 erms invpcid rdseed adx smap clf
lushopt clwb sha_ni xsaveopt xsavec
xgetbv1 xsavec avx_vnni vnni umip wa
itpkg gfni vaes vpclmulqdq rdpid mov
diri movdir64b fsrm md_clear seriali
ze ibt flush_l1d arch_capabilities

Virtualization features:
Virtualization:              VT-x
Hypervisor vendor:           Microsoft
Virtualization type:         full
Caches (sum of all):
L1d:                          288 KiB (6 instances)
L1i:                          192 KiB (6 instances)
L2:                            7.5 MiB (6 instances)
L3:                            12 MiB (1 instance)
NUMA:
NUMA node(s):                1
NUMA node0 CPU(s):          0-11
Vulnerabilities:
Gather data sampling:        Not affected
Ghostwrite:                  Not affected
Indirect target selection:   Not affected
Itlb multihit:               Not affected
L1tf:                         Not affected
Mds:                          Not affected
Meltdown:                    Not affected
Mmio stale data:              Not affected
Old microcode:                Not affected
Reg file data sampling:      Mitigation; Clear Register File
Retbleed:                     Mitigation; Enhanced IBRS
Spec rstack overflow:        Not affected
Spec store bypass:           Mitigation; Speculative Store Bypass
                              disabled via prctl
Spectre v1:                   Mitigation; usercopy/swapgs barriers
                              and __user pointer sanitization
Spectre v2:                   Mitigation; Enhanced / Automatic IBR
                              S; IBPB conditional; PBRSE-eIBRS SW
                              sequence; BHI BHI_DIS_S
Srbds:                        Not affected
Tsa:                          Not affected
Tsx async abort:              Not affected
Vmscape:                      Not affected
varshini@DESKTOP-BLTSL3R:~$
```

```

varshini@DESKTOP-BLT5L3R:~$ ps
  PID TTY          TIME CMD
  1161 pts/0    00:00:00 bash
  1296 pts/0    00:00:00 ps
varshini@DESKTOP-BLT5L3R:~$ top
top - 06:09:01 up 1:23, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 23 total, 1 running, 22 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.
MiB Mem :  7784.6 total,  7142.1 free,   532.0 used,   261.6 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used.  7252.6 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+
1	root	20	0	21904	13116	9720	S	0.0	0.2	0:01.14
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.02
8	root	20	0	3196	2128	2008	S	0.0	0.0	0:00.00
54	root	19	-1	50368	16432	15332	S	0.0	0.2	0:00.64
104	root	20	0	25416	6684	5128	S	0.0	0.1	0:01.22
156	systemd+	20	0	21344	13180	11056	S	0.0	0.2	0:00.15
159	systemd+	20	0	91036	7988	7024	S	0.0	0.1	0:00.27
174	root	20	0	4244	2680	2432	S	0.0	0.0	0:00.02
175	message+	20	0	9608	5416	4692	S	0.0	0.1	0:00.33
191	root	20	0	17980	8836	7792	S	0.0	0.1	0:00.25
195	root	20	0	1755848	11752	9064	S	0.0	0.1	0:00.39
198	syslog	20	0	222516	6024	4616	S	0.0	0.1	0:00.17
215	root	20	0	3124	1976	1836	S	0.0	0.0	0:00.02
233	root	20	0	107032	23252	13796	S	0.0	0.3	0:00.14
312	root	20	0	6704	4664	3876	S	0.0	0.1	0:00.02
358	varshini	20	0	20156	11400	9452	S	0.0	0.1	0:00.14
359	varshini	20	0	21168	3576	1848	S	0.0	0.0	0:00.00
381	varshini	20	0	6080	5324	3660	S	0.0	0.1	0:00.02
717	polkitd	20	0	308172	8092	7196	S	0.0	0.1	0:00.22
1156	root	20	0	3184	1124	980	S	0.0	0.0	0:00.00
1157	root	20	0	3200	1260	1108	S	0.0	0.0	0:00.64
1161	varshini	20	0	6080	5384	3664	S	0.0	0.1	0:00.10
1297	varshini	20	0	9300	5684	3500	R	0.0	0.1	0:00.01